

sions. Although it has been in operation for more than 20 years, the shuttle has continually evolved and is significantly different today than when it first was launched. NASA has made literally thousands of major and minor modifications to the original design that have made it safer, more reliable and more capable today than ever before.

The longest the shuttle has stayed in orbit on any single mission is 17.5 days on mission STS-80 in November 1996. Normally, missions may be planned for anywhere from five to 16 days in duration. The shuttle is designed to reach orbits ranging from about 185 kilometres to 643 kilometres high.

In 2011, NASA finally retired the Space Shuttle fleet after 30 years of service as the space research organisation is focussing more on manned interplanetary missions. The final space shuttle mission, STS-135, ended on July 21, 2011 when Atlantis rolled to a stop at its home port, NASA's Kennedy Space Centre in Florida.

Spacelab

During development of the space shuttle in the 1970's, NASA recognized the need for a facility to allow scientists to conduct experiments while in orbit. Under a cooperation agreement with NASA, the European Space Agency (ESA) built a modular research laboratory that would fit inside the Shuttle's cargo bay, called the Spacelab.

On what was only the ninth Shuttle mission, Spacelab-1 was launched on November 28, 1983. Spacelab was designed to fly in the Space Shuttle cargo bay and enabled scientists to perform a wide range of experiments in microgravity.

The Spacelab system comprised two laboratory modules in short and long versions with airlocks and optical windows, a tunnel and five separate pallets, which could be mounted separate from the Spacelab and exposed experiments to the space environment.

The habitable laboratory module is a room much like the inside of the passenger cabin of a large jet. Three mission and payload specialists can work there simultaneously, surrounded by research and control apparatus in an almost Earth-like environment. They breathe normal air, wear conventional clothes, and set the temperature and humidity to their own liking. Except for weightlessness, they can do their research almost as if they were in any modern high-technology laboratory in their campus, industrial, or government research centres on Earth.

The U-shaped pallet in the rear of the cargo bay serves as a base for research equipment needing a more open view than is feasible from the module, or requiring direct exposure to the radiations and vacuum of outer space.

A pallet weighs 1,200 kilograms, is 4 metres wide and 3 metres long, and can